HOW DOES SIZE MATTER?

INVESTIGATING THE RELATIONSHIPS AMONG PLANT SIZE, INDUSTRIAL STRUCTURE, AND MANUFACTURING PRODUCTIVITY

by

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Abstract

Industrial concentration and market power have been studied extensively at the national scale, in fields ranging from economics and industrial organization to regional science and economic development. At the regional scale, however, industrial structure and firm size relationships have received little attention outside of non-generalizable case studies, primarily because accurate measurements require difficult-to-obtain plant- or firm-level information. Readily available secondary data sources on establishment size distributions (such as *County Business Patterns* or the *Census of Manufactures*) cannot be linked to performance information for particular establishments or firms. Yet region-specific industrial structure may be a crucial determinant of firm performance and thus regional economic fortunes as well (Chinitz 1961; Christopherson and Clark 2007).

This paper examines how industrial concentration and agglomeration economies impact plant performance, focusing on the influence of establishment size in mediating these effects. The *Longitudinal Research Database* of the U.S. Census Bureau is accessed to construct production functions for three manufacturing industries nationwide. These production functions, specified at the establishment level, incorporate characteristics of establishments, industries, and regions, including spatially-differentiated measures of agglomeration economies. Establishment size is evaluated both as an absolute metric and relative to other regional industry plants, as theory suggests that absolute size may be most pertinent to agglomeration benefits but relative size more relevant to industrial structure (Caves and Barton 1990; Bothner 2005). The research builds on earlier work by the author that establishes a direct link between regional industry concentration and the productivity of manufacturing establishments.

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INTRODUCTION

The relationship between industrial structure and economic performance has long interested researchers in a number of fields, including geography, economics, regional science, and economic development. Several aspects of industrial structure, such as average firm or establishment size, industry scale, and patterns of market power and firm interaction, have been studied in depth at the national scale. In a seminal article published in 1961, urban economist Benjamin Chinitz described possible linkages between economic structure at the regional scale and the capacity of a regional economy to perform efficiently at a given moment and also to adapt to changing conditions over time. Employing the contrasting examples of the Pittsburgh and New York metropolitan regions as illustrations, Chinitz examined multiple facets of regional industrial structure, including the absolute size of regional industries, sectoral diversity within the regional economy, and concentration—the extent to which one or a few large firms dominate the regional mix of businesses.

Another aspect of regional industrial structure suggested by Chinitz's article is the degree of dominance or concentration *within* a particular industry in a locality. Regional industrial concentration may be defined as the extent to which the economic activity of a particular industry in a region is accounted for by one or a few large firms. Chinitz's discussion implies that regional concentration may limit the ability of firms and establishments to take advantage of the localized benefits of agglomeration. This research focuses on examining how regional industry-specific concentration affects the performance of the businesses in that regional industry.

Despite a sizable amount of quantitative research stemming from Chinitz's original work, regional-scale industrial concentration has not been investigated systematically across regions other than through non-generalizable case studies. In large part, this omission is because accurate assessment of regional industrial concentration requires firm- or establishment-level information that is typically difficult to obtain. Firms may encompass multiple establishments of varying sizes in a given locality, rendering publicly available secondary data sources on establishment or firm size distributions (such as *County Business Patterns* or the *Census of Manufactures*) inadequate. A second explanation lies in the intricacy of the issues associated with concentration. The precise manner in which spatially proximate firms in an industry affect each other depends on numerous factors, including the nature of the products and customers

associated with the industry, local and regional governance structures, and the degree of extra-regional competition (Okada 2005; Christopherson and Clark 2007; Grabher *et al.* 2008).

Establishment size may affect the ways in which industrial concentration impacts plant performance. The capacity of a plant to gain production advantage from local agglomeration economies may depend upon its size, insofar as plant size reflects both the influence that the firm exerts locally outside the establishment and the resources and organizational complexity internal to the plant. For example, small establishments that support less job specialization and can offer fewer promotion opportunities may have greater susceptibility to employee poaching by a locally dominant company, and thus may experience greater negative effects from regional industrial concentration than larger establishments. In earlier empirical work, establishment size has been investigated with measures of average plant employment or sales that, though obtainable from public data sources, are inadequate for assessing interfirm relationships or plant-specific outcomes. As is the case with the concept of regional industrial concentration, adequate operationalization of establishment size relationships requires plant-level information.

This paper investigates regional industrial concentration in three selected manufacturing industries nationwide, with the focus directed at the role of plant size. Establishment-level data are accessed from the *Longitudinal Research Database* (LRD), a confidential dataset compiled by the United States Census Bureau that contains detailed information on location, industry classification, employment, and inputs and outputs for nearly all manufacturing establishments nationwide. Regional industrial concentration and absolute and relative plant size measures are constructed based on shipment values and employment. Production functions incorporating these measures along with variables indicating potential regional agglomeration economies are estimated at the establishment level, enabling a detailed examination of the relationships among regional industrial concentration, external economies, plant size, and performance.

The research presented here complements earlier analyses that explore and document the extent of regional industrial concentration throughout the United States manufacturing sector and establish linkages between regional industrial concentration and diminished establishment-level productivity within the same set of U.S. manufacturing industries examined in this study (Drucker and Feser 2007; Drucker 2009; 2010). The earlier finding that regional industrial concentration is associated with reduced productivity, particularly for small and medium-sized manufacturing establishments, helps motivate this closer look at the role of plant size. This

paper replicates the methodological approach of Drucker and Feser (2007) in estimating establishment-level productivity, but focuses primarily on the interactions between establishment size and industrial concentration, along with localized agglomeration economies and the outcome of productivity as a measure of economic performance.

There are practical implications to better characterizing the phenomenon of regional concentration or dominance, as industrial structure may be a crucial factor in determining firm performance and overall regional economic fortunes. Industrial dominance is a specific mechanism by which industries may "lock in" to a particular set of competencies (Christopherson and Clark 2007). As markets evolve and technology changes, those competencies, once key economic engines, eventually may become economic liabilities and limit industrial flexibility (Grabher 1993; Bergman 2002). This notion is demonstrated by one of the cases that Chinitz highlighted: steel manufacturing in the metropolitan region Pittsburgh region. Pittsburgh eventually remade its economy around the medical and information technology sectors, but not without a wrenching transition that decimated regional manufacturing employment and required the breakdown of institutional structures and interfirm linkages "hardened" around the steel industry and dominant steel companies (Flammang 1990; Treado and Giarratani 2008). Moreover, industrial concentration may affect adaptability in industries that are not the largest, most visible, or most central in their regional economies. Charlotte, known for banking and finance, has a small and highly concentrated device manufacturing industry; in Buffalo, historically tied to the grain processing, steel, and automobile industries, device manufacturing employment is spread much more evenly among local firms than in Charlotte. As another illustration, plastics and rubber manufacturing jobs are concentrated in several large firms in Des Moines, but the same industry displays a relatively balanced distribution of firm sizes in Toledo. Many such contrasting examples exist in the United States and in other nations, spanning numerous manufacturing industries and regions.

The next section of the paper describes the relationship between industrial concentration and economic performance and reviews previous research pertaining to regional industrial structure and establishment size. After the data sources, measures, and empirical methodology are described, the empirical analysis is presented in three sections: the first focuses on the import of whether individual plants are part of the largest or smallest firms in a regional industry; the second addresses interactions between the firm size classifications and the measures of

regional concentration and agglomeration; and the last segment considers absolute plant size. A brief summary of the findings is offered in the concluding section.

BACKGROUND AND ANTECEDENTS

According to the theory of agglomeration economies, regional industrial concentration may diminish economic performance within regional industries by hampering the ability of firms and establishments to take advantage of the localized benefits of agglomeration. This may occur in a number of ways. Industrial concentration may limit local input pools, as large firms tend to be more vertically integrated and are more likely to purchase inputs from nonlocal suppliers, curtailing accessible markets for specialized suppliers to serve other industry firms (Scott 1986; Scott 1988a; Scott and Kwok 1989; Mason 1991; Enright 1995a; Porter 1998; Henderson *et al.* 2001). Producers of specialized inputs and services favor the stability of large volume contracts and attend first to those purchasers with greater buying power at the expense of smaller firms (Nelson and Winter 1982; Booth 1986). Suppliers serving several different industries may perceive less risk in adapting products for an industry with many rivals than for one dominated by one or a few enterprises. Labor pools may be affected as well. Mobile workers, particularly those possessing specialized training, tend to gravitate toward locally dominant, stable employers (Audretsch 2001), and potential job seekers are more likely to invest in obtaining industry-specific skills in the presence of rivalrous firms (Porter 1990). Public goods and specialized information are more likely to be available or tailored toward particular industry needs in regions in which an industry is competitively structured (Porter 1998; Mukkala 2004). Interfirm networking and group learning opportunities tend to be superior in an environment with many small, independent establishments (Malecki 1994; Carree and Thurik 1999; Gordon and McCann 2000; Helmsing 2001). Bureaucratic management structured to retain control over employees and maximize efficiency tends to inhibit innovation and spin-offs (Booth 1986; Saxenian 1994a), whereas a competitive industrial environment encourages more risk-taking behavior, boosting knowledge spillovers (Scherer 1980; Porter 1990; Malmberg and Maskell 2002). Some of these mechanisms may operate upon firms in related industries as well as in the regionally concentrated industry.

It is also possible that industrial concentration offers some economic advantages. Large firms acting as regional industry anchors may attract specialized labor and intermediate suppliers

to the region, and serve to acclimate local lenders to the particular characteristics of the industry, both of which benefit smaller firms in the industry (Feldman 2003). Established firms generate knowledge and technology spillovers that can provide opportunities for fledgling enterprises (Agrawal and Cockburn 2003). The relative mix of advantage versus disadvantage presented by regional industrial concentration may vary with firm and industrial life cycles as well as the level of maturity of the regional economy (Duranton and Puga 2001).

Some researchers interested in the questions raised by Chinitz have conducted detailed case studies, concentrating on the industrial structures of particular regions or on examining individual regional industries (some examples include Scott 1988b; Saxenian 1994b; Enright 1995b; Gort and Sung 1999; Rantisi 2002; Watts *et al.* 2003). This approach yields a wealth of interesting contextual detail and nuance, but does not provide generalizability across industries, geographic regions, or time periods. Other scholars have performed analyses across regions, focusing on sectoral diversity or average firm or establishment size, structural characteristics that can be assessed using publicly and readily available data. Empirical investigations have found that greater regional diversity and smaller average firm or plant size are generally positively related to a broad range of desirable outcomes, such as firm births, employment growth, economic stability, technology and innovation, productive efficiency, transfer of beneficial spillovers, and regional population growth (Acs and Audretsch 1990; Begovic 1992; Friedman 1995; Harrison *et al.* 1996; Quigley 1998; Feldman and Audretsch 1999; Combes 2000; Hanson 2001; Armington and Acs 2002; Dissart 2003; Henderson 2003; Acs and Armington 2004; Rosenthal and Strange 2004b; Loveridge and Nizalov 2007; Boschma and Iammarino 2009; Glaeser and Kerr 2009). Another group of studies has investigated industry scale, basing the approach in the agglomeration economies literature. The size or scale of an industry is conceptualized as an indicator of localization economies that serve to benefit productivity, profitability, employment growth, and innovation performance, encourage firm births, and attract firms making location choices (Moomaw 1986; Henderson *et al.* 1995; Henderson 1997; Feser 2001a; Desmet and Fafchamps 2005; Kambhampati and McCann 2007).

Analyses of industry scale, sectoral diversity, and average plant or firm size generate useful information and insights, yet these notions are insufficient for studying regional industrial concentration. Industry scale does not consider the internal structure of a regional industry, whereas diversity measures the mix of economic activities in a region rather than industrial

structure. Average firm and establishment size indicate the degree of industrial concentration within a region only in an aggregate fashion. Concentration may have implications for regional economic performance separate from these other features.

Although industrial concentration has not been studied comprehensively at the regional level, there is a body of work within the field of industrial organization that investigates firm sizes and concentration at national scales. This research has centered around Gibrat's Law, the proposition that firm growth rates are independent of firm size. The subject of the firm size distribution is less prominent than many other topics in industrial organization, perhaps due in part to the necessity of using disaggregated data to avoid being subject to aggregation bias (Sutton 1997; Bottazzi and Secchi 2003; Gans and Quiggin 2003). Moreover, most industrial organization research has failed to consider location or spatial variation as a relevant factor, instead emphasizing idiosyncratic or sector-specific explanations of observed industry differences (Schmalensee 1989; Davies and Geroski 1997). A few rare cross-industry studies conducted at the national level provide some evidence that the relationship between industrial concentration and *firm* performance is nonlinear (Caves and Barton 1990; Gopinath *et al.* 2004) and that employment concentration in large firms may boost productivity growth (Acs *et al.* 1999). It is a substantial methodological problem that the prevailing approach—matching precisely defined distributions to empirical phenomena—sets up "extreme" hypotheses that cannot be evaluated with standard inferential statistics (Ijiri and Simon 1977; Powell 2003). The alternative is to measure firm size distributions with summary indicators rather than fitting fully specified distributions. Measures of concentration based on size traits such as employment or sales have been examined extensively in relation to profit rates, productivity, and innovation intensity (Cohen and Levin 1989; Schmalensee 1989; Caves and Barton 1990; Vossen 1999; Gopinath *et al.* 2004). Although this tactic entails complications, as the multiple candidate indicators offer different properties (Needham 1978), it presents a useful approach for this research.

Two empirical studies have come close to the concept at the center of this research by investigating the relative distribution of firm sizes at the regional scale. Rosenthal and Strange (2003) used Dun & Bradstreet's *MarketPlace* database to estimate the distribution of regional employment by plant size category for six industries, reporting that a larger proportion of industry employment in small establishments is associated with more firm births and greater

employment in recently founded establishments. Feser (2002) is both a methodological and conceptual antecedent for this research in that he included a control for manufacturing sector concentration in an establishment-level productivity analysis of spatially attenuating agglomeration economies. He found a strong negative relationship between concentration and productivity in the innovation-intensive measuring and controlling devices industry, but the association is statistically insignificant for the farm and garden equipment industry.

The industrial strategy and ecology literature provides some guidance in terms of the operationalization of plant size in relation to industrial structure. A few studies in this field have argued for the importance of relative size—i.e., in comparison to competitors—as opposed to the more commonly studied absolute size, as a positive influence on business performance outcomes. Bothner (2005) found relative size to have a positive impact on sales growth in the U.S. computer industry. Survival rates of American breweries and of automobile firms in the United States, United Kingdom, France, and Germany are negatively associated with size differences with respect to competitor firms (Hannan *et al.* 1998; Carroll and Swaminathan 2000; Dobrev and Carroll 2003). Although absolute size may be reflective of internal characteristics such as economies of scale, relative size may better represent the influence exercised by the establishment or parent firm within the region. The paucity of empirical work considering relative size may be explained by the lack of available plant-level information, which also leads policymakers and economic development practitioners to focus on absolute employment as a practical gauge of plant and firm size.

METHODOLOGY, DATA, AND VARIABLES

Estimation Approach

This study estimates micro-level production functions in order to test the relationships among regional industrial concentration, agglomeration economies, and productivity. A substantial amount of empirical work investigates agglomeration economies at the regional scale using production functions (see the discussions in Eberts and McMillen 1999; Rosenthal and Strange 2004a). Until about the mid 1990s, such research most often estimated aggregate industry production functions for selected industrial classifications with publicly available regional- or state-level data, despite the problems of ecological fallacy and the lack of industry-specific capital data. Since then, plant-level estimation has largely supplanted the aggregate

approach. Establishment-specific capital information is available from appropriate micro-level records, and aggregation bias is not an issue. Both aggregate and micro-level studies have tended to find substantial positive agglomeration influences on productivity, with variation due to the sources of agglomeration economies, industries, and establishment sizes (e.g., Malmberg *et al.* 2000; Feser 2001a; 2002; Rigby and Essletzbichler 2002; Henderson 2003).

The production function is specified in translog form:

$$\ln Q = \alpha_0 + \sum_i \alpha_i \ln X_i + \frac{1}{2}\sum_i \sum_j \beta_{ij} (\ln X_i \ln X_j) + \sum_k \gamma_k \ln Z_k + \sum_i \sum_k \mathrm{I}(i,k)\lambda_{ik} \ln X_i \ln Z_k + \sum_k \sum_l \mathrm{I}(k,l)\lambda_{kl} \ln Z_k \ln Z_l \quad (1)$$

where Q is establishment output; X represents four conventional inputs (capital, labor, energy, and materials); Z is a vector of regional economic characteristics including industrial concentration and agglomeration economies; i and j index the inputs in X; and k and l index the components of Z. The translog form avoids imposing strong *a priori* assumptions such as constant returns to scale. The two indicator functions permit selective inclusion of interaction terms: the first allows the variables in Z to enter the production function in factor-augmenting form; the second includes the square of concentration and concentration-agglomeration cross-terms as independent variables.

Following the inverse demand function methodology of Kim (1992), cost share equations derived from first-order profit maximization conditions are estimated jointly with the production function in order to improve efficiency. The model system is estimated using iterated nonlinear seemingly unrelated regression. All non-dummy variables are mean-centered to enter the production function. Variables not measured as percentages or ratios are transformed with natural logarithms; the coefficient estimates are interpreted as elasticities at the sample means. A more detailed account of the methodology is available in Drucker and Feser (2007).

Endogeneity is a concern in empirical agglomeration research. Some scholars contend that the firms that are the most productive for unobserved reasons may be those that are best at identifying receptive, productivity-enhancing locations. Such location selectivity would suggest that agglomeration variables are endogenous in the production function, determined jointly with production (Hanson 2001; Henderson 2003; Graham and Kim 2008). The issue is less of a problem here than for many earlier analyses, because the exogeneity of agglomeration advantages is a much more reasonable assumption for individual establishments than for

aggregate industries. The production function contains establishment-specific variables for multiple sources of agglomeration economies, thereby measuring directly relevant plant-level characteristics that were not observable in previous research. Furthermore, recent empirical analyses demonstrate that endogeneity distortions are typically small and are as likely to understate as to exaggerate the influence of agglomeration (Henderson 2003; Rice *et al.* 2006; Koo and Lall 2007; Puga *et al.* 2007; Ellison *et al.* 2010).

Plant-Level Data

The plant-level data source is the *Longitudinal Research Database* (LRD) of the United States Census Bureau. The LRD is compiled from confidential establishment-level records collected for the quinquennial *Census of Manufactures* and the *Annual Survey of Manufactures*, and contains detailed information on establishment locations (by county), primary industry classification, and other establishment characteristics. [See McGuckin (1990) for database construction and content details.]

Although the LRD includes entries for all U.S. establishments reporting under a manufacturing industry code, the coverage of most data items is incomplete for small establishments in non-census years.[1] Since this study focuses on the distributions of large and small establishments, only data from census years are used in order to obtain an accurate balance among establishment sizes. Establishments that are exempt from complete filing requirements (typically those with five or fewer employees) are excluded since they do not report directly on production inputs. The production system is estimated cross-sectionally by industry for each of the three most recent census years available: 1992, 1997, and 2002.

The LRD is compiled from confidential records, and use of the dataset and release of descriptive statistics and results obtained from its analysis are strictly regulated. All of the information presented has been reviewed by Census Bureau staff to ensure confidentiality. Because confidentiality restrictions and disclosure screening requirements limit the types and

[1] The ASM is a five-year panel sample of plants with rotating membership. Only large plants (normally those with at least 250 employees) are included with certainty; the remainder of the sample is selected randomly to reduce data gathering costs and reporting burdens. Sample weights support imputations to national industries or entire manufacturing sectors, but not to industries at the regional scale. In any given year the ASM includes less than 20 percent of United States manufacturing plants. (Thus, despite its name, the LRD is not a longitudinal database at the establishment level.)

quantity of information possible to extract, in some places qualitative descriptions replace numerical tabulations or other quantitative information.

Industries

Three manufacturing industries are examined: plastics and rubber (SIC 30), metalworking machinery (SIC 354), and measuring and controlling devices (SIC 382).[2] Each satisfies several criteria. The industries contain enough establishments spread across regions to present adequate variation in regional industrial concentration and sufficient observations to support the translog estimation system. Industry plants are flexible in location choice and relatively homogeneous in production technology. The selection serves to contrast two traditional industries producing standardized products in a capital-intensive manner with a more technology-intensive industry (measuring devices) subject to shorter innovation cycles.

Regions

The geographic regions are 1990 Labor Market Areas (LMAs) as defined by the United States Department of Agriculture (United States Department of Agriculture 2003). The LMAs are amalgamated from counties to approximate the boundaries of functional economic areas, covering the entire United States. Alaska and Hawaii are excluded due to their relatively isolated locations, and the LMAs covering the urban centers of New York, Los Angeles, and Chicago are omitted as outliers because of those regions' size, density, and volume of international linkages.

Concentration

The concept of *regional* industrial concentration has not appeared in quantitative empirical research so there is no standard measure for it. Previous work conducted at the national scale has adopted a variety of indicators for industrial concentration or market power, including concentration ratios, likelihood ratios, inequality indices, entropy measures, and sample firm size variance (Hay and Morris 1991; Amato 1995; Powell and Lloyd 2005). Empirical comparisons conclude that no single measure is superior. Concentration ratios are the most widely used, in part because they are available from the Census at the national level in public-release versions of the *Census of Manufactures* (Golan *et al.* 1996).

[2] Henceforth the title "measuring and controlling devices" will be shortened to "measuring devices" for brevity.

For this research, the concentration ratio is advantageous in that it is relatively insensitive to small end of the firm size distribution, in accord with the conception of dominance as the degree to which an entire industry is influenced by a few large firms. Therefore, this analysis uses a five-firm concentration ratio. (Drucker and Feser 2007 examine additional measures; the results are qualitatively similar to those presented here.) Establishments are first aggregated to the level of firms based on the same-industry same-region manufacturing components of multi-unit firms.[3] The concentration ratio is the fraction of total regional industry shipment value accounted for by the five largest firms. Because regional industrial concentration is only meaningful in situations in which "dominant" companies can be distinguished from a larger set of potentially impacted firms, regions containing fewer than twelve firms in the industry are excluded.[4]

Establishment Size

Establishment size measures are incorporated into the production model in several iterations. The first approach places two dummy variables in the production function that identify the largest and smallest firms in the regional industry: plants belonging to the five largest ("dominator") firms, and plants belonging to firms reporting less than one tenth the shipment value of the smallest dominator firm ("dominated"). The second iteration adds multiplicative terms interacting the two binary size indicators with the concentration and agglomeration variables. These approaches both concentrate on relative size, so the final iteration substitutes a dummy variable signifying plants employing fewer than a predefined number of workers; three different thresholds are tested.

Additional Independent Variables

This study focuses on regional industrial concentration and plant size; therefore, the remainder of the variables in the production model are described only cursorily here. Appendix A contains the full list of all of the variables along with their data sources and construction, and

[3] For this aggregation step, industries are defined at the most detailed classification level available in the LRD: four-digit SIC or six-digit NAICS industry categories, assigned according to primary production activity. Establishments belonging to multi-unit firms that are located within different regions or that are classified into different industries remain separate "firms". Establishments exempt from full reporting requirements are included here since all establishments report shipment values.

[4] Sensitivity checks demonstrate that varying the minimum number of firms for inclusion from as few as six to as many as fifty alters the strength of the estimations but not the qualitative interpretations of the results obtained.

Drucker and Feser (2007) provides descriptive information and estimation results for the full set of variables in the model.

The production model includes five variables indicating potential local agglomeration economies, designed to be conceptually similar to measures employed successfully in recent research and constructed using information from the LRD combined with publicly available data. One variable indicates labor pooling, two measure supply pooling (manufactured inputs and business services), and two reflect knowledge spillovers (academic research expenditures and private sector patenting). The agglomeration variables are calculated for each establishment, incorporating intraregional spatial attenuation with distance for all but the patenting measure.

Dummy variables for Census Regions proxy different macro-regional levels of development and macroeconomic conditions. Regional unemployment rates and median household income levels capture local economic conditions, and population density helps control for regional size, level of resources, and the absolute dimension of potential agglomeration economies, as well as urban congestion and other agglomeration diseconomies. A regional Herfindahl-Hirschman index calculated across four-digit SIC industries measures regional industrial diversity (inversely), controlling for "Jacobs externalities" that pertain within regions across different industries (Glaeser *et al.* 1992). The change in regional industrial concentration and diversity from twenty years prior to the model year distinguish cumulative or persistent effects.

Descriptives

Table 1 reports the total number of establishments included in the study samples along with the count of those omitted from analysis. Most of the excluded plants are those that are exempt from full filing requirements. The establishments retained in the samples tend to be larger than overall industry averages, a reminder that the analysis does not include the very smallest manufacturers. Plants belonging to regional dominator firms average three to four times the size of the typical plant and ten to thirty times the size of dominated firm plants, depending on whether employment or shipments is used as the barometer of size. Increases in mean establishment sizes and the fraction of plants classified as dominator firms over time reflect consolidation that has accompanied declining employment in the manufacturing sector. Further descriptive information is contained in Drucker and Feser (2007). There are a couple of

Table 1. Samples.

Industry	*Plastics and Rubber*			*Metalworking Machinery*			*Measuring Devices*		
Year	1992	1997	2002	1992	1997	2002	1992	1997	2002
Sample observations	6,747	8,000	6,546	5,189	5,490	4,161	1,384	1,540	1,201
Dropped observations	6,169	6,499	5,128	4,053	4,522	3,982	2,385	2,582	2,211
Percent retained in sample	52.2	55.2	56.1	56.1	54.8	51.1	36.7	37.4	35.2
Mean employment	78	82	91	33	38	36	97	94	111
Mean shipments	9,912	12,789	16,259	3,417	5,191	5,185	12,891	17,603	22,393
Dominator establishments	645	833	901	427	497	505	167	212	202
Percent	9.6	10.4	13.8	8.2	9.1	12.1	12.1	13.8	16.8
Mean employment	286	280	273	148	154	123	410	359	409
Mean shipments	46,714	56,044	60,529	19,014	27,802	22,238	61,399	80,882	92,503
Dominated establishments	3061	3701	2487	2686	2886	1846	658	687	505
Percent	45.4	46.3	38.0	51.8	52.6	44.4	47.5	44.6	42.0
Mean employment	23	24	26	13	15	15	21	23	23
Mean shipments	1,835	2,254	2,835	964	1,462	1,562	1,958	2,800	3,056
Remainder of establishments	3041	3466	3158	2076	2107	1810	559	641	494
Percent	45.1	43.3	48.2	40.0	38.4	43.5	40.4	41.6	41.1
Mean employment	89	97	91	36	41	34	93	82	80
Mean shipments	10,236	13,642	14,199	3,384	4,966	4,122	11,269	12,540	13,491

Note: Value of shipments reported in thousands of nominal dollars.

additional items worth noting here. The samples are size-skewed for each of the three study industries: most plants have input and output quantities below the mean and the dispersion is greater (has a longer tail) at the large end. Regional industry concentration is negatively correlated with most of the agglomeration economy variables, since larger and denser regions provide more potential agglomeration economies and are less likely to be dominated.

EMPIRICAL RESULTS

Relative Size Indicator Variables

Table 2 contains the parameter estimates for the concentration, plant size, and agglomeration variables using the first approach described above of including dummy variables to identify plants in dominator and dominated firms within the regional industry. (Full results along with regression diagnostics and goodness-of-fit measures are available in Drucker and Feser 2007.)

Table 2. Selected Parameter Estimates.

Plastics and Rubber		1992			1997			2002		
Variable	Description	Coeff.	Std. Err.	p Value	Coeff.	Std. Err.	p Value	Coeff.	Std. Err.	p Value
D	concentration	-0.045	0.039	0.25	-0.051	0.033	0.12	-0.065	0.037	0.08
D^2	concentration squared	-0.451	0.259	0.08	-0.301	0.215	0.16	-1.057	0.263	0.00
LP	labor pool	0.900	0.593	0.13	0.040	0.324	0.90	0.686	0.344	0.05
ln SP	inputs supply pool	0.005	0.013	0.67	0.000	0.011	0.98	-0.011	0.013	0.41
ln SD	services supply pool	-0.005	0.012	0.66	0.000	0.012	0.97	0.016	0.013	0.22
ln RS	research	0.002	0.009	0.86	0.007	0.007	0.32	0.005	0.008	0.50
ln PS	patenting	0.003	0.012	0.81	0.020	0.010	0.04	0.021	0.011	0.07
D · LP	conc.-agglom. interactions	0.372	2.754	0.89	-1.368	1.030	0.18	-0.850	1.283	0.51
D · ln SP	conc.-agglom. interactions	0.024	0.061	0.69	0.035	0.045	0.44	0.014	0.053	0.80
D · ln SD	conc.-agglom. interactions	-0.046	0.051	0.37	-0.044	0.043	0.30	-0.106	0.051	0.04
D · ln RS	conc.-agglom. interactions	0.039	0.037	0.29	0.041	0.028	0.14	0.033	0.035	0.34
D · ln PS	conc.-agglom. interactions	-0.061	0.045	0.18	-0.014	0.038	0.72	-0.123	0.040	0.00
D · ln K	conc.-input interactions	0.021	0.004	0.00	0.006	0.003	0.07	0.012	0.005	0.01
D · ln L	conc.-input interactions	0.027	0.008	0.00	-0.003	0.006	0.62	0.023	0.007	0.00
D · ln E	conc.-input interactions	0.001	0.001	0.35	-0.002	0.001	0.01	0.001	0.001	0.46
D · ln M	conc.-input interactions	0.035	0.008	0.00	-0.006	0.006	0.31	0.017	0.008	0.03
DE	dominator firm	0.141	0.013	0.00	0.149	0.012	0.00	0.192	0.013	0.00
SE	dominated firm	-0.191	0.010	0.00	-0.174	0.009	0.00	-0.159	0.010	0.00

Metalworking Machinery		1992			1997			2002		
Variable	Description	Coeff.	Std. Err.	p Value	Coeff.	Std. Err.	p Value	Coeff.	Std. Err.	p Value
D	concentration	-0.088	0.041	0.03	-0.200	0.041	0.00	-0.190	0.052	0.00
D^2	concentration squared	0.287	0.287	0.32	0.821	0.277	0.00	-0.052	0.328	0.87
LP	labor pool	-0.512	0.973	0.60	-2.826	0.936	0.00	0.060	0.630	0.92
ln SP	inputs supply pool	0.024	0.017	0.15	0.030	0.018	0.09	-0.040	0.018	0.03
ln SD	services supply pool	-0.012	0.013	0.36	-0.046	0.016	0.00	0.025	0.017	0.14
ln RS	research	-0.029	0.010	0.00	0.005	0.011	0.65	-0.019	0.011	0.08
ln PS	patenting	0.076	0.017	0.00	0.083	0.015	0.00	0.106	0.017	0.00
D · LP	conc.-agglom. interactions	-1.368	4.734	0.77	-2.749	3.265	0.40	0.801	2.585	0.76
D · ln SP	conc.-agglom. interactions	-0.095	0.080	0.23	0.023	0.085	0.79	-0.099	0.084	0.24
D · ln SD	conc.-agglom. interactions	0.051	0.050	0.30	0.041	0.060	0.49	0.132	0.073	0.07
D · ln RS	conc.-agglom. interactions	0.013	0.037	0.73	-0.040	0.037	0.27	-0.018	0.044	0.68
D · ln PS	conc.-agglom. interactions	0.035	0.076	0.65	0.029	0.057	0.62	-0.121	0.068	0.07
D · ln K	conc.-input interactions	0.004	0.003	0.13	0.008	0.003	0.02	0.019	0.005	0.00
D · ln L	conc.-input interactions	-0.025	0.011	0.03	-0.030	0.010	0.00	0.000	0.012	0.98
D · ln E	conc.-input interactions	-0.001	0.001	0.10	0.001	0.001	0.26	0.001	0.001	0.42
D · ln M	conc.-input interactions	0.023	0.007	0.00	0.028	0.006	0.00	0.031	0.008	0.00
DE	dominator firm	0.178	0.017	0.00	0.210	0.016	0.00	0.217	0.018	0.00
SE	dominated firm	-0.173	0.011	0.00	-0.125	0.011	0.00	-0.158	0.014	0.00

Measuring Devices		1992			1997			2002		
Variable	Description	Coeff.	Std. Err.	p Value	Coeff.	Std. Err.	p Value	Coeff.	Std. Err.	p Value
D	concentration	-0.353	0.183	0.05	-0.250	0.144	0.08	0.118	0.179	0.51
D^2	concentration squared	1.219	0.951	0.20	2.706	1.217	0.03	-3.046	1.720	0.08
LP	labor pool	1.326	0.843	0.12	0.365	0.615	0.55	-0.268	0.889	0.76
ln SP	inputs supply pool	-0.022	0.026	0.40	0.028	0.019	0.13	-0.004	0.022	0.87
ln SD	services supply pool	0.003	0.023	0.90	-0.017	0.018	0.34	-0.017	0.024	0.48
ln RS	research	0.024	0.012	0.04	0.017	0.010	0.09	0.011	0.013	0.40
ln PS	patenting	0.091	0.044	0.04	0.082	0.039	0.04	0.061	0.042	0.15
D · LP	conc.-agglom. interactions	7.862	4.016	0.05	-3.220	3.825	0.40	-6.706	5.826	0.25
D · ln SP	conc.-agglom. interactions	-0.109	0.140	0.44	0.115	0.118	0.33	-0.372	0.182	0.04
D · ln SD	conc.-agglom. interactions	0.071	0.107	0.51	-0.173	0.109	0.11	0.157	0.132	0.24
D · ln RS	conc.-agglom. interactions	-0.013	0.053	0.81	0.058	0.062	0.36	-0.139	0.076	0.07
D · ln PS	conc.-agglom. interactions	0.025	0.275	0.93	0.118	0.262	0.65	0.626	0.305	0.04
D · ln K	conc.-input interactions	0.007	0.007	0.27	-0.003	0.009	0.74	-0.005	0.012	0.65
D · ln L	conc.-input interactions	0.064	0.022	0.00	0.035	0.021	0.10	0.024	0.027	0.39
D · ln E	conc.-input interactions	-0.004	0.001	0.01	-0.005	0.002	0.00	-0.002	0.001	0.08
D · ln M	conc.-input interactions	0.036	0.018	0.04	0.024	0.019	0.22	-0.004	0.024	0.88
DE	dominator firm	0.231	0.035	0.00	0.251	0.032	0.00	0.275	0.036	0.00
SE	dominated firm	-0.272	0.029	0.00	-0.254	0.026	0.00	-0.222	0.031	0.00

Regional industrial concentration has a significant negative influence on productivity at the plant level. Recalling that the coefficients are elasticities, in the plastics and rubber industry, a 20 percent rise in the five-firm concentration ratio of industry shipment value in an LMA (approximately one standard deviation) is associated with a decline of 0.9 to 1.3 percent in output at the sample means of the other variables and holding all else constant, depending on the year of the cross-section. The effects are larger in the other two industries: a rise of 20 percent in the total industry shipment value accounted for by the top five firms in an LMA relates to a 1.8 to 4.0 percent drop in output in metalworking machinery plants, and a 7.1 decline in production in 1992 and a 5.0 percent drop in 1997 in measuring devices establishments. The estimated coefficient of concentration is positive but insignificant in the 2002 measuring devices cross-section.

The two firm size variables are very strongly associated with output. In all nine of the industry-year cross-sections, dominator establishments outperform, and dominated plants underperform, the sample averages by substantial margins. Depending on the particular industry-year combination, dominator plants demonstrate a productivity advantage of 14 to 28 percent, and dominated plants suffer a production deficit of similar magnitude, compared to the remainder of the regional industry establishments that are neither part of dominator nor dominated firms. In each model, firm size is the strongest influence on output other than input quantities. These impacts—both the direction and the scale—make sense. Dominator firms tend to have more resources at their disposal, greater local influence, and can take advantage of economies of scale, whereas dominated firms have access to fewer resources and economies of scale than the average establishment.

Because the production function includes the two indicator variables denoting plants belonging to the largest and smallest firms in each regional industry, the coefficients of the regional industrial concentration measure do not simply reflect dominator firms outperforming locally dominated enterprises. The estimates described above refer to the effect of concentration on output accounting for relative firm size. In other words, regional industrial concentration influences plant production independently of its status as part of a dominator or a dominated firm.

The agglomeration variables demonstrate mixed influence on production. The benefits of potential regional labor pools are small and vary by year for each industry. The estimated

coefficients of the two supply pooling variables are rarely large enough to be important, and are of opposite sign in most of the cross-sections, suggesting that colinearity obscures the impacts on output. Knowledge spillovers show more substantial impacts. Proximity to relevant academic research has a notable productivity benefit, but only for plants in the measuring devices industry. The relatively small influence of academic research is not unexpected, as basic research impacts productivity mainly in the very long term and research expenditures track only one of many means by which universities influence economic performance. The effects of private sector patenting are positive and large in all three study industries. Doubling the regional patent rate in relevant technology fields is associated with up to two percent higher output for plastics and rubber establishments, an eight to ten percent increase for metalworking machinery manufacturers, and a six to nine percent output rise in measuring device plants. Even considering the smallest of these estimates, the influence of private sector innovation is strong enough to reveal a route by which policy can influence productivity.

The estimated coefficients of the concentration variable indicate direct effects evaluated at the sample means of the independent variables. Because the regional industrial concentration variable enters the production equation nonlinearly, with a quadratic term and interactions with both the conventional input and agglomeration economy variables, the estimated marginal effects of concentration vary with these interacted terms, permitting simple modeling of nonlinear effects. For example, a large coefficient for the square of concentration (reported in the measuring the plastics and rubber industry in all three years and in the measuring devices industry in 2002) yields increasingly negative impacts of concentration on performance increases as the concentration ratio rises. The opposite occurs in 1992 and 1997 for the metalworking machinery and the measuring devices industries: the positive concentration-squared term indicates that the effects of concentration on production are strongest in those regions with moderate levels of industrial concentration.

The interaction terms between industrial concentration and the four standard production inputs—capital, labor, energy, and materials—provide another way to investigate the role of plant size in mediating the influence of concentration on establishment-level productivity. The vertical axes in Figure 1 display the estimated marginal impacts of regional industrial concentration on output—the percent change in output associated with doubling the concentration ratio from the sample mean with all other variables held constant. The horizontal

axes contain six points that describe the range from low to high quantities of the conventional inputs. Disclosure restrictions preclude percentiles; using hypothetical input combinations upholds U.S. Census Bureau confidentiality requirements. The point labeled "mean" represents the sample means of the four conventional inputs. Points A through E are percentages of the sample means constructed as proportions of the means (chosen separately for each input across each of the nine cross-sectional samples) to approximate the observed sample range. A, B, and C are smaller than the sample means and D and E are larger but they are not necessarily spaced equally.

This approach differs from the analysis of the two firm size dummy variables in that it infers size from inputs rather than output (shipment values), categorizes size at the establishment rather than the firm level, and examines hypothetical input combinations rather than data pertaining to individual plants or firms. Because the points are constructed along the multivariate range of input quantities, relative size remains the focus, yet the two firm size dummy variables still are taken into account. In effect, Figure 1 reveals the degree to which the establishment size differentials that remain after accounting for the major distinction among dominant, dominated, and remaining firms affect the influence of concentration on plant production. The point of estimation remains the sample means, so that the estimations become less reliable moving toward the extremes of the ranges, as reflected by the confidence intervals. Therefore, the variation in the effects of concentration is best interpreted qualitatively.

In the plastics and rubber industry, the effects of regional industrial concentration are small but negative at the sample means. In 1997, as plant size shifts away from the mean amounts of the four conventional inputs, the impact of concentration changes only slightly while the significance of the estimate declines (the confidence intervals widen and include zero). In 1992 and 2002, however, small plants experience greater and more significantly negative effects of concentration. In other words, concentration acts as more of a hindrance to productivity performance for smaller plastics and rubber plants in the region in 1992 and 2002. The largest establishments even benefit from industry-specific concentration in their regions. The latter pattern is replicated in most of the graphs in Figure 1: the marginal effect of concentration is greater in magnitude (a larger negative number) and more significant for smaller plants. Only the 1992 metalworking machinery and the 2002 measuring devices models display a result more like 1997 plastics and rubber, wherein the effect of concentration is relatively stable across

establishment sizes. Industry-specific conditions in those particular years may have favored the smaller firms in heavily concentrated regional industries.

The estimated impacts of regional industrial concentration vary much less with the agglomeration regime than they do with input quantities. The concentration-agglomeration interaction terms in Table 1 are predominantly small and vary in sign in non-systematic fashion. Figure 2 is constructed like Figure 1 but with horizontal axes representing hypothetical regional endowments of agglomeration economies. In the 2002 cross-section for the plastics and rubber and measuring devices industries, concentration has positive productivity effects in regions with few available agglomeration economies and negative effects where the levels of agglomeration economies are large. Industrial concentration may hinder establishments from accessing localized agglomeration economies, lowering productivity in regions with greater potential agglomeration benefits. In most of the cross-sections, however, there is little change in concentration with different levels of agglomeration economies. This outcome suggests that agglomeration economies are not the primary mechanism by which regional dominance reduces productivity. Concentration appears to diminish establishment-level productivity, but not by restricting the ability of regional manufacturers to access local benefits of agglomeration.

Firm Size Interactions

The production function is easily modified by adding interaction terms between the two firm size dummy variables and the regional industrial concentration and agglomeration variables. The marginal impacts of regional industrial concentration and agglomeration economies are estimated for each of the three size groups: plants within dominant firms, plants within dominated firms, and plants that do not belong to either category. This approach deepens the examination of the conditioning effects that firm size has on the influences of concentration and agglomeration on output.

With multiple size categories and nine industry-year cross-sections, the revised model yields a large volume of numerical results. Table 3 summarizes the marginal effects of regional industrial concentration, the five agglomeration economies, and the firm size dummies, with all

Figure 1. Marginal Impacts of Regional Industrial Concentration By Conventional Inputs Ranges.

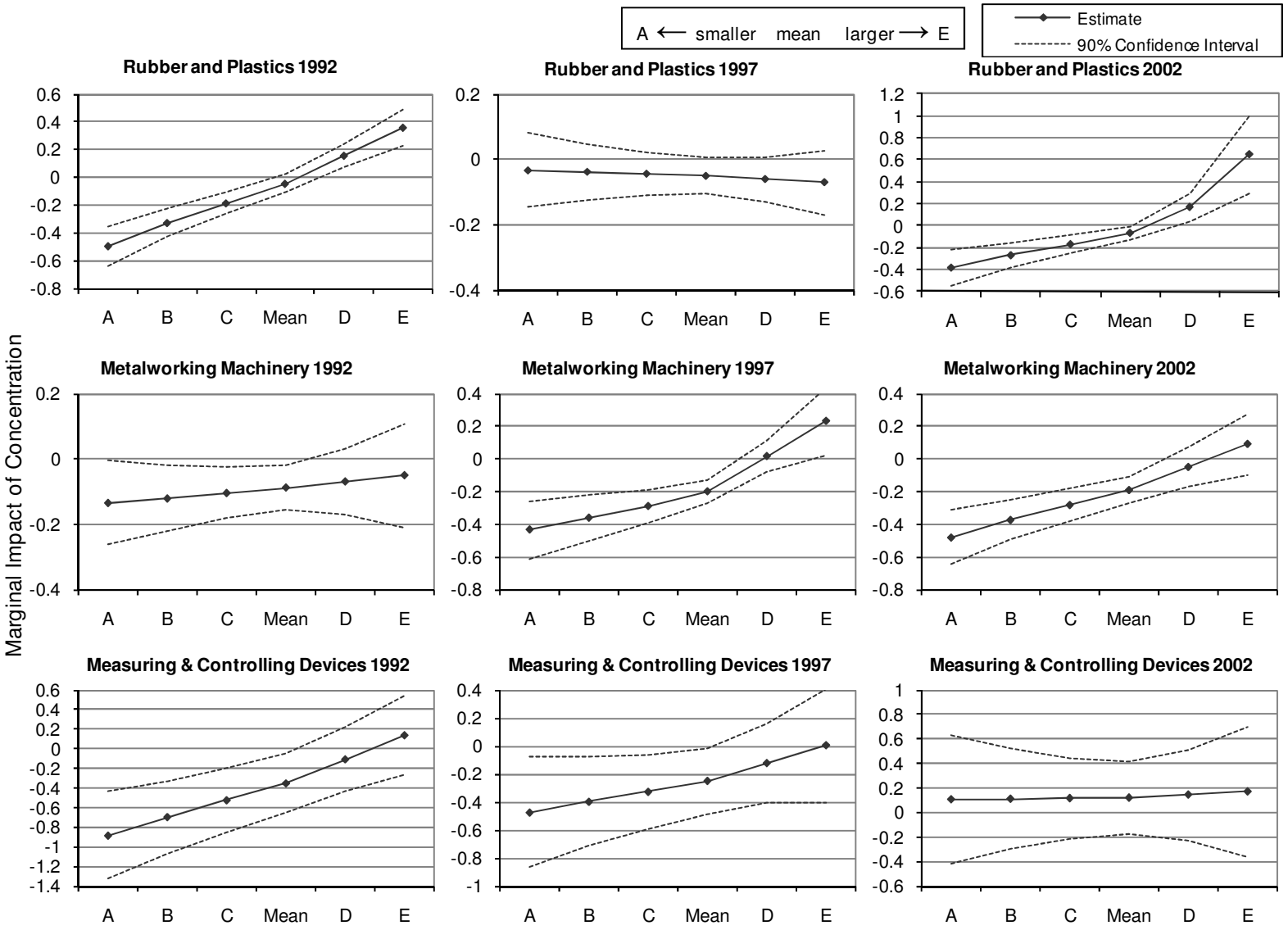
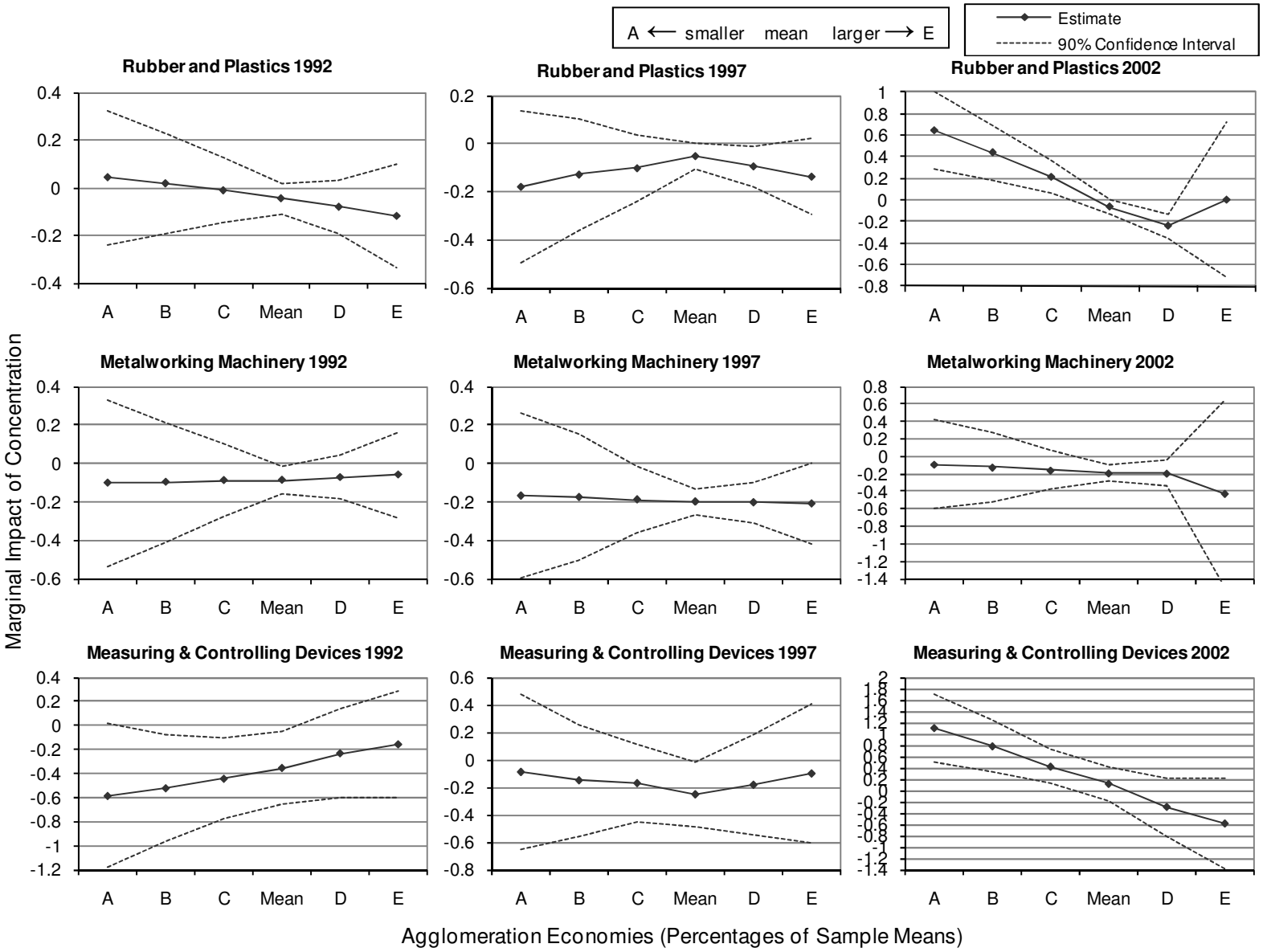


Figure 2. Marginal Impacts of Regional Industrial Concentration By Agglomeration Economies Ranges.



other variables held at their sample means, presenting only the signs and estimated significance ranges of the effects. (Appendix B contains the full marginal impacts.) Since the significances pertain to interaction terms signifying marginal effects, they coincide with magnitude and serve as a normalized measure of influence strength. The size classifications subdivide the estimation samples whereas the model maintains the complexity of the translog production function, so that conventional significance levels are more difficult to obtain than in the first approach. Therefore, Table 3 (and Table 6 below) reports marginal effects that are significant at the 80 percent confidence level or greater.

Table 3 reveals that the influence of regional industrial concentration on production tends to be the most negative for those establishments in the plastics and rubber and the metalworking machinery industries that belong to neither dominator nor dominated parent firms. In the measuring devices samples, dominated plants rather than those in the "neither dominator nor dominated" category are the most negatively affected by concentration. Few patterns with respect to the agglomeration measures are consistent in holding their significance and direction over time. Metalworking machinery plants tend to exhibit a negative influence on production from research, particularly with regard to the "neither dominator nor dominated" relative size classification. The highly substantial positive effect of regional patenting on the productivity of metalworking machinery establishments is not differentiated by firm size category.

Absolute Plant Size Interactions

The approaches taken so far concentrate on relative size, but plant size may be measured either relative to competitor enterprises or in absolute terms. Some studies adopt relative size measures, such as Feser (2001b), in which size categories are defined by sample quartiles (see *Background and Antecedents* for additional examples). Absolute size classifications are more common in research applications, however, and are used nearly exclusively in policy settings, because constructing relative size measures requires information about the entire set of plants or firms in the sample or population. It may be reasonable on theoretical grounds to postulate that relative size affects the influence of regional industrial concentration on production whereas absolute size is more pertinent to the benefits to be gained from localized agglomeration economies, but there is no previous direct empirical evidence available as to whether this supposition holds in practice.

Table 3. Firm Size Interaction Marginal Effects.

			Plastics and Rubber			*Metalworking Machinery*			*Measuring Devices*		
Variable	Description	Interaction	1992	1997	2002	1992	1997	2002	1992	1997	2002
D	Concentration	dominator	++					−	−		
		neither	−−−	−−−	−−−	−−−	−−−	−−−	−−		
		dominated				−−	−−−	−−	−−−	−−−	
LP	Labor Pool	dominator		−−		+++	−−−	+			−
		neither			+++		−−−				
		dominated	+				−−−				
SP	Inputs Supply	dominator					++	−−		+++	
	Pool	neither	+			+		−−−		+	
		dominated					+				
SD	Services	dominator					−−	+++		−−	−
	Supply Pool	neither			++		−	++			
		dominated					−−−				
RS	Research	dominator			+	−−		−			
		neither		+++		−−−		−−		+	
		dominated				−−					
PS	Patenting	dominator				+++	+++	+++			+++
		neither	−		+	+++	+++	+++		++	
		dominated		+++	++	+	+++	+++		+	

Notes: A single plus or minus sign indicates significance at the 80 percent confidence level, a double sign 90 percent confidence level, and a triple sign 95 percent confidence level.

The final methodological strategy replaces the two firm size variables that indicate relative size with a single dummy variable identifying plants employing fewer than a predefined number of workers. The marginal effects of regional industrial concentration and agglomeration economies may be computed separately for "small" and "not small" establishments. There are several reasons to test multiple thresholds for determining which establishments are small. There is substantial variety in definitions of small businesses across various policies and nations (Table 4).[5] The way in which establishment scale conditions the influence of concentration and

[5] Within the United States, the small business size standards of the Small Business Administration are industry-specific, but for most manufacturing industries the criterion is that a small establishment employs no more than 500 full-time equivalent workers (United States Small Business Administration 2006). For compliance purposes, the Environmental Protection Agency considers small businesses to be those with a maximum of 100 employees (United States Environmental Protection Agency 2000). The Small Business Job Protection Act of 1996 defines small businesses as having 100 or fewer employees for the purpose of establishing employee savings options, whereas the maximum size for companies to gain exemption from the 1993 Family and Medical Leave Act is 49 workers (United States Department of Labor 2002; n.d.). The European Commission and the United Kingdom categorize enterprises with fewer than 50 employees as small and those with less than 250 employees as medium-sized (European Commission 2005; United Kingdom Department for Business Enterprise and Regulatory Reform 2005). Smaller nations not surprisingly tend to maintain smaller maximum sizes for small business classifications. For example, the threshold for eligibility for government small business programs is 19 or fewer employees in Australia and New Zealand (New Zealand Ministry of Economic Development 2005; Australia Department of

Table 4. Selected Establishment Size Definitions.

Organization / Nation	Purpose / Coverage	Definition
U.S. Small Business Administration	manufacturing	0-500 employees
U.S. Environmental Protection Agency	environmental compliance	0-100 employees
U.S. Small Business Job Protection Act (1996)	exemption from Family and Medical Leave Act	0-49 employees
	employee savings options	0-100 employees
European Commission	all industries	0-49 employees
United Kingdom	all industries	0-49 employees
Australia	all industries	0-19 employees
New Zealand	all industries	0-19 employees

Note: for sources, see footnote 5.

agglomeration opportunities on productivity may differ according to the size threshold considered. In addition, altering the definition may yield a sense of the robustness of the results obtained. Three different employment thresholds are evaluated: 15, 50, and 250 employees.[6] (Two additional cutoff levels—100 and 500 employees—were also tested, generating results that follow the same general patterns.)

Table 5 presents sample descriptive information about the absolute size categories. Very few plants in any of the three study industries qualify as other than small when the criterion is to employ more than 250 workers. In part, this is due to measuring employment at the establishment rather than the firm level, but the main reason is that the study industries have highly skewed plant size distributions (as do most industries). The great majority of businesses fall within the definitions of "small" used in many policy contexts. Small business applicability thresholds are often set high in order to maximize the set of firms included in a program or subject to a set of guidelines. Even when the threshold size is reduced to 50 employees, more than 80 percent of the metalworking machinery establishments are "small", as are some 60 percent of the plastics and rubber and measuring devices plants. Despite the fact that 15

Industry Tourism and Resources 2007). Note that these are illustrative examples that do not cover the breadth of policies and definitions found in the United States or worldwide.

[6] These are the number of employees, both full-time and part-time, reported by each establishment as part of the *Census of Manufactures*. The exclusion of establishments exempt from complete filing requirements removes the very smallest stand-alone enterprises, typically those with five or fewer employees.

Table 5. Absolute Size Classifications – Sample Information.

Industry	*Plastics and Rubber*			*Metalworking Machinery*			*Measuring Devices*		
Year	1992	1997	2002	1992	1997	2002	1992	1997	2002
Sample observations	6,747	8,000	6,546	5,189	5,490	4,161	1,384	1,540	1,201
Mean employment	78	82	91	33	38	36	97	94	111
Small (250 or fewer employees)	6,351	7,474	6,081	5,110	5,397	4,101	1,271	1,422	1,088
Percent	94	93	93	98	98	99	92	92	91
Small (50 or fewer employees)	4,037	4,650	3,552	4,459	4,585	3,482	886	969	718
Percent	60	58	54	86	84	84	64	63	60
Small (15 or fewer employees)	1,415	1,688	1,099	2,717	2,603	1,921	362	376	299
Percent	21	21	17	52	47	46	26	24	25

employees is considerably smaller than most absolute size definitions used in the United States, nearly half of the metalworking machinery plants and between a fifth and a quarter of the establishments in the other two study industries meet the criterion.

Table 6 is constructed the same way as Table 3 for the absolute size classification approach, presenting only the signs and estimated significance ranges of the model coefficients. (Appendix B contains the full marginal impacts.) Starting with the dummy variables taken by themselves, productivity differentials are not as clear cut for absolute as for relative size. Small plastics and rubber establishments are less productive, all else equal, when small is defined so as to include all plants up to 250 employees, but are more productive on average than other establishments when the small designation covers only plants with 15 or fewer workers. If the absolute size threshold is set at 100 employees, the overall productivity difference between small and large is insubstantial. The pattern does not hold as precisely for the other two study industries, but the same general conclusion is supported: very small size is advantageous, but when defined more widely, such as is common in policy definitions, small establishments are at a productivity disadvantage in comparison with the other plants in the industry. One possible explanation for the productivity advantage of the smallest plants is that they have assets that cannot be gauged in this analysis, such as links to parent firms, proprietary production processes, or differentiated products (i.e., they may more commonly function as specialized or boutique manufacturers rather than competitive mass producers).

Although the direct relationship of size to production is highly significant for both the

Table 6. Absolute Plant Size Interactions.

Variable	Description	Interaction	1992			1997			2002		
Plastics and Rubber			Sm.≤250	Sm.≤50	Sm.≤15	Sm.≤250	Sm.≤50	Sm.≤15	Sm.≤250	Sm.≤50	Sm.≤15
SM	Small		−−−		+++	−−−		+++	−−−		+++
D	Concentration	large	+++						+++	+	+
		small			−						
LP	Labor Pool	large			+++					+	+
		small	++	+							
SP	Inputs Supply	large									
	Pool	small									
SD	Services	large									+
	Supply Pool	small							+		
RS	Research	large							−		
		small					−−	−			
PS	Patenting	large				+++			+++	+	+
		small									
Metalworking Machinery			Sm.≤250	Sm.≤50	Sm.≤15	Sm.≤250	Sm.≤50	Sm.≤15	Sm.≤250	Sm.≤50	Sm.≤15
SM	Small			−−−		−−−	−−−	+++	−	−−−	+++
D	Concentration	large	+++	+							
		small				−−−	−−−	−−−	−	−	−−
LP	Labor Pool	large			+	−−−		−			
		small				−−−	−−	−−−			
SP	Inputs Supply	large				+++					−−−
	Pool	small							−−−	−−−	−
SD	Services	large				−−−					+++
	Supply Pool	small	+	+				−−		+	
RS	Research	large	−	−−−					+	−	
		small									
PS	Patenting	large					++	+++		+++	+++
		small				++	+		+++	++	
Measuring Devices			Sm.≤250	Sm.≤50	Sm.≤15	Sm.≤250	Sm.≤50	Sm.≤15	Sm.≤250	Sm.≤50	Sm.≤15
SM	Small			−−−	−−−	−−−		++	−		+
D	Concentration	large									
		small		−	−−			−−			
LP	Labor Pool	large	+++								
		small	+++		+++	+	+				
SP	Inputs Supply	large	−−−						−		−
	Pool	small	−						−	−	−
SD	Services	large	++								
	Supply Pool	small									
RS	Research	large			++		++	++			
		small	++			++					
PS	Patenting	large		+		+	++		+		
		small			−						

Notes: A single plus or minus sign indicates significance at the 80 percent confidence level, a double sign 90 percent confidence level, and a triple sign 95 percent confidence level.

absolute and relative size variables, the magnitude of the absolute size influence is much smaller in most cases. For example, in the 1992 plastics and rubber sample, small plants with 15 or fewer employees produce approximately eight percent more output on average than other establishments. The output of establishments with up to 250 employees averages four percent less than larger plants. In contrast, plants belonging to dominator firms produce nearly 14 percent greater output and plants belonging to dominated firms produce 19 percent less than establishments that are neither dominators nor dominated (Table 2). Similar comparisons hold for the other years and industries. With respect to total output, the size status of a manufacturing firm relative to other regional establishments in the industry carries more influence than absolute plant size. The marginal effects of regional industrial concentration are more clearly influenced by the relative than the absolute size classifications as well, as indicated by the stronger and more consistent patterns contained in the *Concentration* rows of Tables 3 compared to Table 6. These findings are consistent with the proposition that relative size better represents the influence within a region exerted by an establishment or its parent firm.

Turning to the interactions, small establishments usually are negatively affected by regional industrial concentration, or equivalently large establishments are positively influenced, but none of the size definitions tested produces a noticeably stronger demarcation of the conditioning effect of absolute size. Many of the calculated marginal effects fail to reach even the 80 percent confidence level threshold. As with relative size, few patterns emerge for the agglomeration variables. For example, labor pooling possibilities seem to favor both large and small measuring devices establishments in 1992, but provide almost no significant productivity advantages for any size plants within the industry in 1997 or 2002. The influence of labor pooling is almost uniformly negative in the metalworking machinery industry in the 1997 model, for both small and large plants, but the effect is not present in 1992 or 2002. Plastics and rubber plants that employ between 15 and 250 workers seem to benefit from locally available labor pools only in 1992. The two supply pooling variables exhibit inconsistent behavior with respect to establishment size as well, not a surprising outcome given the ambiguity in their direct effects. The two knowledge spillovers demonstrate conditioning influences that fit with the direct effects observed earlier. In 1992 and 1997, measuring device manufacturers benefit from proximity to related academic research, particularly those of intermediate size: smaller than 250 but greater

than 15 employees. Patenting has a highly significant positive effect on productivity for metalworking machinery establishments in the same intermediate size range.

Summary and Conclusions

This study demonstrates the importance of regional industrial concentration and business size. Regional industrial concentration has a limiting effect on the productivity of manufacturing plants in the three industries studied. The productivity of large, dominant firms is higher on average, and the productivity of the smallest, most dominated firms is lower on average, than for other establishments in the regional industry. Small plants, as measured by the number of workers, tend to have lower productivity than large plants, though the tendency reverses for the very smallest establishments with fewer than 15 employees. The negative effects of regional industrial concentration on productivity are enhanced for smaller and less dominant plants.

None of the three thresholds used to define "small" in terms of absolute establishments size is superior to the others in terms of revealing meaningful interactions, yet the differences observed between small and large plants certainly are not robust with respect to the definition of "small". In other words, it matters where the threshold is set. Relative size is more influential than absolute size in conditioning the impact of regional industrial concentration on production, supporting the idea that external regional influence, rather than organizational or other characteristics internal to the establishment or firm, is key to this relationship.

Local sources of agglomeration economies seem not to be a primary mechanism by which regional industrial dominance and concentration affect production. Nor does establishment size have consistent influences on agglomeration. This may reflect the weakness of some of the agglomeration variables in the model in impacting plant-level productivity.

The primary finding that regional industrial concentration is linked to diminished business productivity, with the relationship stronger for smaller and less locally-dominant businesses, suggests an area of concern for economic development policy. The growth and dynamism of small businesses is a crucial component of regional economic adjustment capability. Industry-specific dominance within a region hinders fledgling businesses, leaving the local area with less flexibility to adapt successfully to changes in economic conditions or to restructure in the face of a major economic disturbance. This study suggests the need for further research to clarify the mechanism by which concentration and business size affect production, in

order to guide the development of effective policies to address issues arising from disadvantageous regional industrial structure.

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APPENDIX A: PRODUCTION FUNCTION VARIABLES AND CONSTRUCTION

Table A-1. Production Function Variables and Sources.

Category	Variable		Description	Source
dependent variable	Q	output	value of output	LRD
standard inputs	K	capital	gross book assets plus (capitalized) rentals	LRD
	L	labor	production-worker-equivalent hours	LRD
	E	energy	value of electricity and fuels	LRD
	M	materials	value of purchased materials and services	LRD
concentration	D	concentration	percent of shipments in dominator firms	LRD
agglomeration economies	LP	labor pooling	percent of local employment in top industry occupations	Census, BLS
	SP	inputs pooling	local employment in input supply industries	LBD, BEA
	SD	producer services	local employment in producer service industries	LBD, BEA
	RS	research	university research expenditures in industry-relevant fields	NSF
	PS	patents	per capita industry-relevant patent rate	USPTO
establishment size	DE	dominator	dummy: establishment belongs to dominant firm	LRD
	SE	dominated	dummy: shipments less than 10 percent of smallest dominator	LRD
	SM	small plant	dummy: employment less than threshold amount	LRD
controls	CR1	region	dummy: South Census Region	Census
	CR2	region	dummy: Midwest Census Region	Census
	CR3	region	dummy: West Census Region	Census
	POP	population	population density (persons per square mile)	Census
	UE	unemployment	unemployment rate	BLS
	INC	income	median household income in region (nominal dollars)	Census
	DV	diversity	Herfindahl index of regional industrial diversity (concentration)	LBD
	DH	historic dominance	dominance in earlier year	LRD
	DVH	historic diversity	diversity (concentration) in earlier year	LBD

LRD = Longitudinal Research Database, LBD = Longitudinal Business Database, BLS = Bureau of Labor Statistics (Staffing Patterns Matrix), BEA = Bureau of Economic Analysis (*Benchmark Input-Output Accounts*: Make and Use tables), NSF = National Science Foundation (CASPAR), USPTO = U.S. Patent and Trademark Office (Utility Patent Reports), Census = U.S. Bureau of the Census (decennial censes, population estimates, *Small Area Income and Poverty Estimates*).

Units: Monetary values are thousands of nominal dollars. Time measures are thousands of hours.

Inputs and Output: Variables are calculated at the establishment level. Output (Q) is the total value of shipments adjusted for inventories and work in progress:

$$Q = TVS + (WIE - WIB) + (FIE - FIB).$$

TVS is the total value of shipments; *WIE* and *WIB* are work in progress at the end and beginning of the year; *FIE* and *FIB* are end-of-year and beginning-of-year finished product inventories.

Capital is constructed as a gross stock measure—the sum of end-of-year capital assets and capitalized building and machinery rental expenditures:

$$K = TAE + \frac{BR}{BPR} + \frac{MR}{MPR}.$$

TAE is total end-of-year building and machinery assets; *BR* is building rental expenditures; *MR* is machinery rental expenditures; *BPR* and *MPR* are (unpublished) annual national 3-digit-SIC capital prices obtained from the Bureau of Labor Statistics averaged across asset categories.

Labor (*L*) is production-worker-equivalent hours, derived as the ratio of total wages to production worker average hourly wages:

$$L = \frac{WP + WNP}{(WP / PH)}.$$

WP and *WNP* are production and non-production payrolls; *PH* is production worker hours.

Energy (*E*) is the sum of purchased electricity and fuel quantities:

$$E = CF\left(\frac{1{,}000}{EPR}\right) + PE\left(\frac{3{,}412.705}{1{,}000}\right).$$

PE is millions of British Thermal Units of purchased electricity; *CF* is purchased fuels cost; *EPR* is the average state-level industry-specific energy price from the U.S. Energy Information Administration *State Energy Data System*.

Materials (*M*) is the total cost of materials, parts, contract work, resales, purchased services, and repairs, adjusted for changes in materials inventories:

$$M = CP + CR + CW + CPC + RB + RM + (MIB - MIE).$$

CP is materials and parts cost; *CR* is expenditures for resales; *CW* is contract work costs; *CPC* is purchased communications services; *RB* and *RM* are building and machinery repairs; *MIB – MIE* is the difference between beginning- and end-of-year materials inventories.

Capital costs are total capital assets adjusted by multiplying by (unpublished) Bureau of Labor Statistics capital asset prices and LRD-reported building and machinery rental costs. Labor costs are total wages and salaries plus supplemental labor expenditures. Energy costs are purchased electricity and fuels costs. Materials costs are identical to the quantity measure.

Agglomeration Economies: Labor pooling is measured as an establishment's access to workers with skills that roughly match the industry's expected occupational requirements:

$$LP_{kx} = \sum_{c}\left(\frac{O_{cx}}{O_{cT}} d_{ck}^{-\alpha}\right)$$

where x is the study industry, c indexes counties, k is the target establishment's county, O_{cx} is county c's residential workforce in the top 15 occupations employed by industry x nationally (identified from the Bureau of Labor Statistics national staffing matrix). O_{cT} is county c's total workforce, and d_{ck} is the distance between the centroids of county c and the county of the target establishment for distances up to 75 miles and zero otherwise. Values of O_c and O_{Tc} are from the 1990 and 2000 Census Equal Employment Opportunity tabulations.

Potential supply pools of manufactured inputs and producer services are calculated by weighting the local presence of supplier industries by the importance of each industry as a supplier to the study industry at the national level. Manufacturing input supply pooling is:

$$SP_{kx} = \sum_c \left(\left(\sum_m \frac{E_{cm} P_{xm}}{P_{xM}} \right) \cdot d_{ck}^{-\alpha} \right)$$

where m indexes manufacturing industries, E_{cm} is county c's employment in industry m, P_{xm} is the dollar amount that the study industry purchases nationally from supplier industry m, P_{xM} is the study industry's total national purchases from the manufacturing sector, and the other notations are the same as for labor pooling. Producer services pooling has nearly the same formula except that purchases and local employment refer to suppliers of producer services:

$$SD_{kx} = \sum_c \left(\left(\sum_s \frac{E_{cs} P_{xs}}{P_{xS}} \right) \cdot d_{ck}^{-\alpha} \right)$$

where s indexes producer services industries and P_{xS} is the study industry's total national purchases of producer services. Purchase amounts are derived from the Make and Use tables of the *Benchmark Input-Output Accounts* from the Bureau of Economic Analysis. E_{cm} and E_{cs} are tabulated from the *Longitudinal Business Database*, another confidential establishment-level Census dataset that covers most economic sectors but lacks full input and output information.

Two variables indicate different types of knowledge spillovers. The first gauges regional access to relevant basic research and knowledge:

$$RS_{kx} = \sum_c \left(\left(\sum_f R_{cf} \right) \cdot d_{ck}^{-\alpha} \right)$$

where f indexes industry-relevant academic fields, R_{cf} is the total amount of research expenditures in academic field f during the previous five years at research universities located in county c, and the other variables are as described above. The maximum distance is set at 200

miles. The academic fields relevant to each industry are modified from Cohen *et al*. (2002), and annual university research expenditures are from the National Science Foundation's CASPAR database. The second measure uses patents to indicate private sector research activity and regional innovative culture, weighting the volume of patents in each technology classification by the relative importance of those technology categories to the target industry:

$$PS_{rx} = \sum_{g \in K} \left(\frac{PAT_{gr}}{POP_r} N_{gx} \right)$$

where g indexes patent technology classifications, r signifies the region, x represents the study industry, K is the set of patent technology classifications relevant to the study industry (both directly and via cross-industry spillovers), PAT_{gr} is the number of utility patents granted within region r in the last five years in patent technology class g (from the CASSIS bibliographic system of the U.S. Patent and Trademark Office), POP_r is the regional residential population, and N_{gx} is a relevance index developed from citation flows (Koo 2005). Industry-relevant patent technology classifications are identified from the USPTO technology-industry match and a technology flow matrix developed by Koo (2005).

Control Variables: Regional unemployment rates, household incomes, and population densities are constructed for LMAs by combining county-level estimates. Unemployment rates are from the Bureau of Labor Statistics *Local Area Unemployment Statistics*. Median household income levels are from Census *Small Area Income and Poverty Estimates*. Population density is calculated from Census population estimates. Establishment-level data from the LBD are used to construct regional industrial diversity and historic diversity measures. Because the LBD starts in 1977, the historic diversity measure for the 1992 sample is from 1977.

APPENDIX B: MARGINAL IMPACT ESTIMATES FOR SIZE INTERACTION MODELS

Table B-1. Plastics and Rubber Firm and Plant Size Interaction Model Estimates.

Plastics and Rubber			Dominance Categories				Small ≤ 250 Employees				Small ≤ 50 Employees				Small ≤ 15 Employees			
Variable	Description	Interaction	Coeff.	Std. Err.	t Stat.	p Value	Coeff.	Std. Err.	t Stat.	p Value	Coeff.	Std. Err.	t Stat.	p Value	Coeff.	Std. Err.	t Stat.	p Value
Year: 1992																		
DE	Dominators		0.1389	0.0171	8.14	0.00												
SE / SM	Dominated / Small		-0.1879	0.0116	-16.19	0.00	-0.0415	0.0209	-1.98	0.05	0.0034	0.0119	0.29	0.77	0.0752	0.0132	5.71	0.00
D	Concentration	dominator	0.1274	0.0686	1.86	0.06												
		neither / large	-0.0923	0.0457	-2.02	0.04	0.2187	0.1028	2.13	0.03	0.0056	0.0498	0.11	0.91	-0.0154	0.0421	-0.37	0.71
		dominated / small	-0.0225	0.0546	-0.41	0.68	-0.0480	0.0409	-1.17	0.24	-0.0531	0.0463	-1.15	0.25	-0.1001	0.0667	-1.50	0.13
LP	Labor	dominator	-0.3656	1.3993	-0.26	0.79												
	Pooling	neither / large	0.8273	0.7739	1.07	0.29	0.0902	2.0237	0.04	0.96	1.0326	0.8463	1.22	0.22	1.2888	0.6561	1.96	0.05
		dominated / small	1.3223	0.8186	1.62	0.11	1.1143	0.6233	1.79	0.07	1.1678	0.7253	1.61	0.11	0.1987	1.1220	0.18	0.86
SP	Inputs Supply	dominator	0.0342	0.0327	1.05	0.30												
	Pool	neither / large	0.0241	0.0175	1.38	0.17	0.0319	0.0478	0.67	0.50	-0.0066	0.0194	-0.34	0.73	-0.0162	0.0144	-1.12	0.26
		dominated / small	-0.0113	0.0165	-0.69	0.49	-0.0142	0.0134	-1.05	0.29	-0.0168	0.0152	-1.11	0.27	-0.0041	0.0221	-0.18	0.85
SD	Services	dominator	-0.0157	0.0258	-0.61	0.54												
	Supply Pool	neither / large	-0.0120	0.0152	-0.79	0.43	-0.0087	0.0380	-0.23	0.82	0.0075	0.0162	0.46	0.65	0.0073	0.0130	0.56	0.57
		dominated / small	0.0035	0.0156	0.22	0.82	0.0053	0.0125	0.42	0.67	0.0055	0.0143	0.38	0.70	-0.0048	0.0210	-0.23	0.82
RS	Research	dominator	-0.0080	0.0182	-0.44	0.66												
		neither / large	-0.0058	0.0110	-0.53	0.60	-0.0167	0.0252	-0.66	0.51	-0.0013	0.0120	-0.11	0.91	0.0023	0.0098	0.23	0.81
		dominated / small	0.0093	0.0111	0.84	0.40	0.0014	0.0094	0.15	0.88	0.0017	0.0102	0.17	0.87	-0.0014	0.0136	-0.10	0.92
PS	Patents	dominator	0.0279	0.0231	1.21	0.23												
		neither / large	-0.0201	0.0151	-1.33	0.18	0.0144	0.0351	0.41	0.68	-0.0107	0.0162	-0.66	0.51	-0.0066	0.0133	-0.50	0.62
		dominated / small	0.0167	0.0160	1.04	0.30	-0.0083	0.0127	-0.66	0.51	-0.0047	0.0143	-0.33	0.74	-0.0093	0.0205	-0.46	0.65
Year: 1997																		
DE	Dominators		0.1524	0.0153	9.95	0.00												
SE / SM	Dominated / Small		-0.1783	0.0107	-16.66	0.00	-0.0643	0.0190	-3.38	0.00	0.0084	0.0107	0.78	0.43	0.0758	0.0124	6.13	0.00
D	Concentration	dominator	-0.0641	0.0553	-1.16	0.25												
		neither / large	-0.1249	0.0402	-3.11	0.00	0.0800	0.0852	0.94	0.35	0.0103	0.0423	0.24	0.81	0.0175	0.0358	0.49	0.62
		dominated / small	0.0166	0.0466	0.36	0.72	0.0197	0.0347	0.57	0.57	0.0333	0.0395	0.84	0.40	0.0189	0.0559	0.34	0.74
LP	Labor	dominator	-1.0979	0.5820	-1.89	0.06												
	Pooling	neither / large	-0.2573	0.3852	-0.67	0.50	-0.8546	0.7782	-1.10	0.27	-0.2714	0.3959	-0.69	0.49	-0.3765	0.3459	-1.09	0.28
		dominated / small	0.4773	0.4005	1.19	0.23	-0.2565	0.3365	-0.76	0.45	-0.2469	0.3754	-0.66	0.51	0.0641	0.5108	0.13	0.90
SP	Inputs Supply	dominator	0.0229	0.0221	1.04	0.30												
	Pool	neither / large	0.0098	0.0145	0.67	0.50	-0.0322	0.0329	-0.98	0.33	-0.0142	0.0152	-0.93	0.35	0.0022	0.0122	0.18	0.86
		dominated / small	-0.0080	0.0144	-0.56	0.58	0.0053	0.0116	0.45	0.65	0.0139	0.0131	1.07	0.29	0.0080	0.0189	0.43	0.67
SD	Services	dominator	-0.0062	0.0221	-0.28	0.78												
	Supply Pool	neither / large	-0.0174	0.0152	-1.15	0.25	-0.0035	0.0328	-0.11	0.92	0.0039	0.0157	0.25	0.80	-0.0027	0.0129	-0.21	0.84
		dominated / small	0.0120	0.0154	0.78	0.44	-0.0007	0.0123	-0.06	0.95	-0.0023	0.0140	-0.17	0.87	0.0052	0.0201	0.26	0.79
RS	Research	dominator	0.0096	0.0133	0.72	0.47												
		neither / large	0.0189	0.0083	2.28	0.02	0.0097	0.0206	0.47	0.64	0.0088	0.0092	0.95	0.34	-0.0015	0.0073	-0.20	0.84
		dominated / small	-0.0031	0.0086	-0.36	0.72	-0.0043	0.0068	-0.63	0.53	-0.0127	0.0075	-1.71	0.09	-0.0159	0.0100	-1.60	0.11
PS	Patents	dominator	0.0089	0.0179	0.50	0.62												
		neither / large	0.0067	0.0124	0.54	0.59	0.0685	0.0272	2.52	0.01	0.0043	0.0128	0.33	0.74	0.0086	0.0107	0.81	0.42
		dominated / small	0.0342	0.0128	2.68	0.01	0.0048	0.0103	0.46	0.64	0.0109	0.0116	0.94	0.35	0.0027	0.0166	0.16	0.87
Year: 2002																		
DE	Dominators		0.2073	0.0165	12.60	0.00												
SE / SM	Dominated / Small		-0.1695	0.0128	-13.25	0.00	-0.1221	0.0224	-5.45	0.00	-0.0030	0.0127	-0.24	0.81	0.1367	0.0155	8.82	0.00
D	Concentration	dominator	0.0245	0.0597	0.41	0.68												
		neither / large	-0.1484	0.0439	-3.38	0.00	0.3485	0.0942	3.70	0.00	0.0767	0.0470	1.63	0.10	0.0525	0.0392	1.34	0.18
		dominated / small	0.0188	0.0612	0.31	0.76	0.0143	0.0386	0.37	0.71	0.0164	0.0451	0.36	0.72	0.0184	0.0747	0.25	0.81
LP	Labor	dominator	-0.3009	0.6117	-0.49	0.62												
	Pooling	neither / large	1.2067	0.4153	2.91	0.00	-0.5022	0.8354	-0.60	0.55	0.6857	0.4283	1.60	0.11	0.5007	0.3679	1.36	0.17
		dominated / small	0.1340	0.4592	0.29	0.77	0.4252	0.3610	1.18	0.24	0.0571	0.4157	0.14	0.89	-0.5207	0.6198	-0.84	0.40
SP	Inputs Supply	dominator	0.0199	0.0269	0.74	0.46												
	Pool	neither / large	-0.0124	0.0170	-0.73	0.46	0.0133	0.0423	0.32	0.75	-0.0113	0.0182	-0.62	0.54	-0.0100	0.0142	-0.70	0.48
		dominated / small	-0.0089	0.0180	-0.49	0.62	-0.0126	0.0133	-0.95	0.34	-0.0081	0.0157	-0.52	0.61	-0.0109	0.0250	-0.44	0.66
SD	Services	dominator	-0.0164	0.0249	-0.66	0.51												
	Supply Pool	neither / large	0.0293	0.0171	1.71	0.09	0.0119	0.0393	0.30	0.76	0.0243	0.0180	1.35	0.18	0.0210	0.0146	1.44	0.15
		dominated / small	0.0016	0.0192	0.08	0.93	0.0189	0.0141	1.35	0.18	0.0089	0.0166	0.54	0.59	-0.0042	0.0262	-0.16	0.87
RS	Research	dominator	0.0229	0.0164	1.39	0.16												
		neither / large	0.0047	0.0100	0.47	0.64	-0.0340	0.0261	-1.31	0.19	-0.0082	0.0109	-0.75	0.45	-0.0087	0.0089	-0.98	0.32
		dominated / small	0.0038	0.0116	0.33	0.74	-0.0050	0.0085	-0.59	0.56	-0.0051	0.0099	-0.51	0.61	0.0082	0.0152	0.54	0.59
PS	Patents	dominator	0.0192	0.0189	1.02	0.31												
		neither / large	0.0178	0.0134	1.33	0.18	0.1025	0.0293	3.50	0.00	0.0225	0.0140	1.61	0.11	0.0188	0.0119	1.57	0.12
		dominated / small	0.0273	0.0161	1.70	0.09	0.0073	0.0117	0.63	0.53	0.0045	0.0137	0.33	0.74	-0.0041	0.0213	-0.19	0.85

Note: "Neither" and "dominated" label the dominance models, "large" and "small" pertain to the small establishment models.

Table B-2. Metalworking Machinery Firm and Plant Size Interaction Model Estimates.

Metalworking Machinery			Dominance Categories				Small ≤ 250 Employees				Small ≤ 50 Employees				Small ≤ 15 Employees			
Variable	Description	Interaction	Coeff.	Std. Err.	t Stat.	p Value	Coeff.	Std. Err.	t Stat.	p Value	Coeff.	Std. Err.	t Stat.	p Value	Coeff.	Std. Err.	t Stat.	p Value
Year: 1992																		
DE	Dominators		0.2139	0.0223	9.60	0.00												
SE / SM	Dominated / Small		-0.1984	0.0154	-12.84	0.00	-0.0411	0.0492	-0.84	0.40	-0.0879	0.0193	-4.55	0.00	0.0132	0.0140	0.94	0.35
D	Concentration	dominator	0.0724	0.0761	0.95	0.34												
		neither / large	-0.1224	0.0484	-2.53	0.01	0.4446	0.2063	2.16	0.03	0.1059	0.0790	1.34	0.18	0.0112	0.0523	0.21	0.83
		dominated / small	-0.1210	0.0663	-1.83	0.07	-0.0174	0.0427	-0.41	0.68	-0.0331	0.0444	-0.75	0.46	-0.0422	0.0518	-0.81	0.42
LP	Labor	dominator	4.7249	2.3084	2.05	0.04												
	Pooling	neither / large	-0.4848	1.2325	-0.39	0.69	-0.6734	5.7337	-0.12	0.91	1.5686	1.9355	0.81	0.42	1.8349	1.2188	1.51	0.13
		dominated / small	-0.4572	1.3373	-0.34	0.73	0.7503	1.0059	0.75	0.46	0.4334	1.0497	0.41	0.68	-0.6534	1.2484	-0.52	0.60
SP	Inputs Supply	dominator	-0.0164	0.0370	-0.44	0.66												
	Pool	neither / large	0.0333	0.0209	1.60	0.11	0.0132	0.0851	0.16	0.88	0.0369	0.0308	1.20	0.23	-0.0078	0.0206	-0.38	0.70
		dominated / small	0.0166	0.0219	0.76	0.45	-0.0141	0.0176	-0.80	0.42	-0.0148	0.0182	-0.81	0.42	-0.0091	0.0211	-0.43	0.67
SD	Services	dominator	0.0105	0.0247	0.43	0.67												
	Supply Pool	neither / large	-0.0136	0.0154	-0.88	0.38	0.0160	0.0707	0.23	0.82	-0.0037	0.0227	-0.16	0.87	0.0161	0.0152	1.06	0.29
		dominated / small	-0.0060	0.0156	-0.39	0.70	0.0182	0.0131	1.39	0.17	0.0177	0.0134	1.32	0.19	0.0163	0.0150	1.09	0.28
RS	Research	dominator	-0.0364	0.0193	-1.89	0.06												
		neither / large	-0.0417	0.0114	-3.65	0.00	-0.0872	0.0539	-1.62	0.11	-0.0435	0.0192	-2.26	0.02	-0.0123	0.0122	-1.01	0.31
		dominated / small	-0.0219	0.0128	-1.72	0.09	-0.0070	0.0100	-0.70	0.48	-0.0053	0.0101	-0.53	0.60	-0.0069	0.0112	-0.62	0.54
PS	Patents	dominator	0.0974	0.0335	2.91	0.00												
		neither / large	0.1076	0.0204	5.28	0.00	0.0449	0.0986	0.46	0.65	0.0146	0.0327	0.45	0.66	0.0037	0.0207	0.18	0.86
		dominated / small	0.0378	0.0230	1.64	0.10	-0.0025	0.0167	-0.15	0.88	-0.0052	0.0172	-0.30	0.76	-0.0131	0.0197	-0.66	0.51
Year: 1997																		
DE	Dominators		0.2254	0.0199	11.30	0.00												
SE / SM	Dominated / Small		-0.1470	0.0142	-10.38	0.00	-0.1750	0.0470	-3.72	0.00	-0.0351	0.0172	-2.04	0.04	0.0306	0.0132	2.32	0.02
D	Concentration	dominator	0.0244	0.0764	0.32	0.75												
		neither / large	-0.1649	0.0487	-3.39	0.00	-0.1251	0.1763	-0.71	0.48	0.0840	0.0682	1.23	0.22	-0.0400	0.0479	-0.83	0.40
		dominated / small	-0.3104	0.0585	-5.30	0.00	-0.1101	0.0416	-2.65	0.01	-0.1383	0.0434	-3.19	0.00	-0.1804	0.0501	-3.60	0.00
LP	Labor	dominator	-3.6442	1.7737	-2.05	0.04												
	Pooling	neither / large	-2.2124	1.0843	-2.04	0.04	-11.0570	4.5390	-2.44	0.01	-1.8475	1.5569	-1.19	0.24	-1.4533	1.0792	-1.35	0.18
		dominated / small	-3.5986	1.1913	-3.02	0.00	-1.8869	0.9587	-1.97	0.05	-1.8421	0.9950	-1.85	0.06	-2.7240	1.1534	-2.36	0.02
SP	Inputs Supply	dominator	0.0750	0.0422	1.78	0.08												
	Pool	neither / large	0.0154	0.0224	0.68	0.49	0.2356	0.1156	2.04	0.04	0.0090	0.0356	0.25	0.80	-0.0010	0.0222	-0.05	0.96
		dominated / small	0.0346	0.0242	1.43	0.15	0.0033	0.0181	0.18	0.85	0.0079	0.0191	0.42	0.68	0.0211	0.0231	0.91	0.36
SD	Services	dominator	-0.0590	0.0333	-1.77	0.08												
	Supply Pool	neither / large	-0.0254	0.0193	-1.32	0.19	-0.2409	0.0895	-2.69	0.01	-0.0224	0.0295	-0.76	0.45	-0.0078	0.0192	-0.41	0.68
		dominated / small	-0.0637	0.0211	-3.01	0.00	-0.0166	0.0162	-1.02	0.31	-0.0168	0.0169	-1.00	0.32	-0.0339	0.0201	-1.69	0.09
RS	Research	dominator	-0.0131	0.0200	-0.66	0.51												
		neither / large	0.0017	0.0127	0.13	0.89	-0.0072	0.0491	-0.15	0.88	-0.0093	0.0184	-0.51	0.61	0.0027	0.0127	0.21	0.83
		dominated / small	0.0108	0.0130	0.83	0.41	0.0047	0.0109	0.43	0.67	0.0040	0.0112	0.36	0.72	0.0042	0.0123	0.35	0.73
PS	Patents	dominator	0.0959	0.0288	3.33	0.00												
		neither / large	0.1032	0.0179	5.77	0.00	-0.0592	0.0842	-0.70	0.48	0.0457	0.0261	1.75	0.08	0.0378	0.0174	2.18	0.03
		dominated / small	0.0671	0.0192	3.49	0.00	0.0257	0.0145	1.78	0.08	0.0202	0.0152	1.34	0.18	0.0107	0.0176	0.61	0.54
Year: 2002																		
DE	Dominators		0.1785	0.0238	7.49	0.00												
SE / SM	Dominated / Small		-0.1649	0.0197	-8.37	0.00	-0.0865	0.0613	-1.41	0.16	-0.0627	0.0228	-2.75	0.01	0.0505	0.0173	2.91	0.00
D	Concentration	dominator	-0.1127	0.0839	-1.34	0.18												
		neither / large	-0.2016	0.0605	-3.34	0.00	0.1612	0.2153	0.75	0.45	-0.0621	0.0990	-0.63	0.53	-0.0340	0.0641	-0.53	0.60
		dominated / small	-0.1906	0.1039	-1.84	0.07	-0.0752	0.0532	-1.41	0.16	-0.0726	0.0552	-1.32	0.19	-0.1145	0.0660	-1.73	0.08
LP	Labor	dominator	1.7215	1.1977	1.44	0.15												
	Pooling	neither / large	0.1833	0.7385	0.25	0.80	-2.4017	4.4641	-0.54	0.59	-0.3869	1.1214	-0.35	0.73	0.9169	0.7586	1.21	0.23
		dominated / small	-0.4248	0.9386	-0.45	0.65	0.0918	0.6503	0.14	0.89	0.2242	0.6781	0.33	0.74	-0.7844	0.8104	-0.97	0.33
SP	Inputs Supply	dominator	-0.0807	0.0417	-1.93	0.05												
	Pool	neither / large	-0.0474	0.0232	-2.05	0.04	-0.1448	0.1389	-1.04	0.30	0.0105	0.0382	0.28	0.78	-0.0524	0.0229	-2.29	0.02
		dominated / small	-0.0278	0.0272	-1.02	0.31	-0.0433	0.0187	-2.32	0.02	-0.0501	0.0196	-2.56	0.01	-0.0382	0.0249	-1.54	0.12
SD	Services	dominator	0.1027	0.0348	2.95	0.00												
	Supply Pool	neither / large	0.0385	0.0208	1.85	0.06	0.0703	0.1347	0.52	0.60	0.0130	0.0322	0.41	0.68	0.0448	0.0205	2.19	0.03
		dominated / small	0.0015	0.0234	0.07	0.95	0.0215	0.0176	1.23	0.22	0.0238	0.0182	1.31	0.19	0.0014	0.0221	0.06	0.95
RS	Research	dominator	-0.0293	0.0214	-1.37	0.17												
		neither / large	-0.0263	0.0136	-1.94	0.05	0.0890	0.0657	1.35	0.18	-0.0297	0.0218	-1.37	0.17	-0.0148	0.0139	-1.06	0.29
		dominated / small	-0.0121	0.0158	-0.76	0.45	-0.0048	0.0114	-0.42	0.67	-0.0007	0.0117	-0.06	0.95	0.0086	0.0137	0.63	0.53
PS	Patents	dominator	0.1279	0.0308	4.16	0.00												
		neither / large	0.1027	0.0208	4.93	0.00	0.0726	0.1082	0.67	0.50	0.0677	0.0324	2.09	0.04	0.0579	0.0210	2.76	0.01
		dominated / small	0.1107	0.0260	4.25	0.00	0.0361	0.0173	2.08	0.04	0.0305	0.0180	1.70	0.09	0.0150	0.0213	0.70	0.48

Note: "Neither" and "dominated" label the dominance models, "large" and "small" pertain to the small establishment models.

Table B-3. Measuring Devices Firm and Plant Size Interaction Model Estimates.

Measuring Devices			Dominance Categories				Small ≤ 250 Employees				Small ≤ 50 Employees				Small ≤ 15 Employees			
Variable	Description	Interaction	Coeff.	Std. Err.	t Stat.	p Value	Coeff.	Std. Err.	t Stat.	p Value	Coeff.	Std. Err.	t Stat.	p Value	Coeff.	Std. Err.	t Stat.	p Value
Year: 1992																		
DE	Dominators		0.2256	0.0514	4.39	0.00												
SE / SM	Dominated / Small		-0.3221	0.0393	-8.19	0.00	-0.0535	0.0580	-0.92	0.36	-0.1577	0.0403	-3.91	0.00	-0.0860	0.0437	-1.97	0.05
D	Concentration	dominator	-0.4002	0.2614	-1.53	0.13												
		neither / large	-0.3731	0.2149	-1.74	0.08	-0.2907	0.3164	-0.92	0.36	-0.1736	0.2315	-0.75	0.45	-0.1940	0.2236	-0.87	0.39
		dominated / small	-0.5132	0.2333	-2.20	0.03	-0.1695	0.1931	-0.88	0.38	-0.2934	0.2119	-1.38	0.17	-0.4761	0.2549	-1.87	0.06
LP	Labor	dominator	0.5425	2.1364	0.25	0.80												
	Pooling	neither / large	0.8845	1.1551	0.77	0.44	6.1252	2.9753	2.06	0.04	1.2334	1.3056	0.94	0.34	0.7206	1.0801	0.67	0.50
		dominated / small	0.5153	1.2268	0.42	0.67	2.0907	0.8883	2.35	0.02	1.2975	1.0662	1.22	0.22	3.4284	1.5833	2.17	0.03
SP	Inputs Supply	dominator	-0.0249	0.0609	-0.41	0.68												
	Pool	neither / large	0.0361	0.0354	1.02	0.31	-0.1948	0.0865	-2.25	0.02	-0.0388	0.0423	-0.92	0.36	-0.0408	0.0346	-1.18	0.24
		dominated / small	0.0216	0.0401	0.54	0.59	-0.0395	0.0281	-1.40	0.16	-0.0414	0.0330	-1.25	0.21	-0.0563	0.0475	-1.19	0.24
SD	Services	dominator	0.0007	0.0472	0.01	0.99												
	Supply Pool	neither / large	-0.0133	0.0299	-0.45	0.66	0.1189	0.0657	1.81	0.07	0.0266	0.0343	0.78	0.44	0.0288	0.0290	1.00	0.32
		dominated / small	-0.0107	0.0339	-0.32	0.75	0.0200	0.0241	0.83	0.41	0.0303	0.0284	1.07	0.29	0.0178	0.0413	0.43	0.67
RS	Research	dominator	0.0157	0.0302	0.52	0.60												
		neither / large	0.0063	0.0166	0.38	0.70	0.0371	0.0359	1.03	0.30	0.0193	0.0199	0.97	0.33	0.0262	0.0155	1.68	0.09
		dominated / small	0.0089	0.0175	0.51	0.61	0.0220	0.0126	1.75	0.08	0.0136	0.0147	0.92	0.36	0.0071	0.0223	0.32	0.75
PS	Patents	dominator	0.0689	0.1032	0.67	0.50												
		neither / large	0.0724	0.0643	1.13	0.26	0.1171	0.1124	1.04	0.30	0.1015	0.0641	1.58	0.11	0.0409	0.0547	0.75	0.45
		dominated / small	0.0188	0.0617	0.30	0.76	0.0308	0.0474	0.65	0.52	0.0081	0.0579	0.14	0.89	-0.1274	0.0890	-1.43	0.15
Year: 1997																		
DE	Dominators		0.2338	0.0453	5.16	0.00												
SE / SM	Dominated / Small		-0.3102	0.0355	-8.73	0.00	-0.1392	0.0596	-2.33	0.02	0.0441	0.0355	1.24	0.21	0.0701	0.0368	1.91	0.06
D	Concentration	dominator	-0.2194	0.2116	-1.04	0.30												
		neither / large	-0.1369	0.1660	-0.82	0.41	0.0509	0.3208	0.16	0.87	0.1055	0.1833	0.58	0.56	0.0840	0.1570	0.53	0.59
		dominated / small	-0.5347	0.2038	-2.62	0.01	-0.0320	0.1510	-0.21	0.83	-0.1519	0.1591	-0.95	0.34	-0.3804	0.2032	-1.87	0.06
LP	Labor	dominator	-0.6915	1.4030	-0.49	0.62												
	Pooling	neither / large	0.3999	0.8070	0.50	0.62	1.1181	2.1853	0.51	0.61	0.0652	0.9406	0.07	0.94	0.6430	0.6974	0.92	0.36
		dominated / small	0.7921	0.8830	0.90	0.37	0.8433	0.6447	1.31	0.19	1.1624	0.7234	1.61	0.11	0.9978	1.0687	0.93	0.35
SP	Inputs Supply	dominator	0.0831	0.0375	2.21	0.03												
	Pool	neither / large	0.0389	0.0242	1.61	0.11	0.0095	0.0688	0.14	0.89	-0.0038	0.0295	-0.13	0.90	-0.0081	0.0217	-0.37	0.71
		dominated / small	-0.0022	0.0302	-0.07	0.94	-0.0055	0.0197	-0.28	0.78	-0.0006	0.0223	-0.03	0.98	0.0103	0.0328	0.31	0.75
SD	Services	dominator	-0.0632	0.0350	-1.81	0.07												
	Supply Pool	neither / large	-0.0085	0.0233	-0.36	0.72	-0.0172	0.0542	-0.32	0.75	0.0183	0.0264	0.69	0.49	0.0113	0.0207	0.54	0.59
		dominated / small	-0.0021	0.0283	-0.08	0.94	0.0120	0.0194	0.62	0.53	0.0015	0.0224	0.07	0.95	0.0079	0.0329	0.24	0.81
RS	Research	dominator	0.0190	0.0240	0.79	0.43												
		neither / large	0.0213	0.0139	1.53	0.13	-0.0097	0.0360	-0.27	0.79	0.0290	0.0166	1.75	0.08	0.0220	0.0121	1.82	0.07
		dominated / small	0.0109	0.0158	0.69	0.49	0.0208	0.0110	1.89	0.06	0.0118	0.0127	0.93	0.35	0.0082	0.0191	0.43	0.67
PS	Patents	dominator	0.0527	0.0780	0.68	0.50												
		neither / large	0.1021	0.0531	1.92	0.05	0.1902	0.1210	1.57	0.12	0.0890	0.0522	1.71	0.09	0.0419	0.0432	0.97	0.33
		dominated / small	0.0687	0.0512	1.34	0.18	0.0116	0.0412	0.28	0.78	-0.0359	0.0475	-0.76	0.45	-0.0497	0.0664	-0.75	0.45
Year: 2002																		
DE	Dominators		0.2789	0.0472	5.91	0.00												
SE / SM	Dominated / Small		-0.1849	0.0392	-4.72	0.00	-0.1025	0.0623	-1.65	0.10	0.0149	0.0387	0.38	0.70	0.0641	0.0424	1.51	0.13
D	Concentration	dominator	0.0995	0.2436	0.41	0.68												
		neither / large	0.1667	0.1906	0.87	0.38	-0.3288	0.3592	-0.92	0.36	0.0722	0.2182	0.33	0.74	-0.0432	0.1980	-0.22	0.83
		dominated / small	0.1824	0.2624	0.70	0.49	0.0544	0.1884	0.29	0.77	0.0218	0.1959	0.11	0.91	0.0556	0.2254	0.25	0.81
LP	Labor	dominator	-2.6861	1.6870	-1.59	0.11												
	Pooling	neither / large	-1.1657	1.2206	-0.96	0.34	0.8088	2.2595	0.36	0.72	0.6876	1.1926	0.58	0.56	0.8094	0.9749	0.83	0.41
		dominated / small	0.4882	1.1892	0.41	0.68	0.7021	0.9332	0.75	0.45	1.0152	1.0423	0.97	0.33	1.2422	1.4418	0.86	0.39
SP	Inputs Supply	dominator	0.0096	0.0413	0.23	0.82												
	Pool	neither / large	-0.0131	0.0295	-0.44	0.66	-0.0910	0.0637	-1.43	0.15	-0.0306	0.0333	-0.92	0.36	-0.0334	0.0256	-1.30	0.19
		dominated / small	0.0136	0.0357	0.38	0.70	-0.0321	0.0236	-1.36	0.17	-0.0377	0.0268	-1.41	0.16	-0.0546	0.0374	-1.46	0.14
SD	Services	dominator	-0.0672	0.0441	-1.52	0.13												
	Supply Pool	neither / large	-0.0090	0.0310	-0.29	0.77	-0.0301	0.0615	-0.49	0.62	0.0012	0.0324	0.04	0.97	0.0199	0.0263	0.76	0.45
		dominated / small	-0.0395	0.0346	-1.14	0.25	0.0188	0.0251	0.75	0.45	0.0245	0.0289	0.85	0.40	0.0083	0.0409	0.20	0.84
RS	Research	dominator	0.0118	0.0249	0.47	0.64												
		neither / large	0.0199	0.0178	1.12	0.26	0.0066	0.0326	0.20	0.84	0.0180	0.0183	0.99	0.32	0.0069	0.0148	0.46	0.64
		dominated / small	0.0042	0.0181	0.23	0.82	0.0107	0.0140	0.76	0.44	0.0036	0.0159	0.23	0.82	0.0233	0.0216	1.08	0.28
PS	Patents	dominator	0.2442	0.0776	3.15	0.00												
		neither / large	0.0632	0.0531	1.19	0.23	0.1877	0.1154	1.63	0.10	0.0401	0.0559	0.72	0.47	0.0232	0.0468	0.50	0.62
		dominated / small	0.0033	0.0616	0.05	0.96	0.0111	0.0444	0.25	0.80	0.0139	0.0500	0.28	0.78	0.0420	0.0667	0.63	0.53

Note: "Neither" and "dominated" label the dominance models, "large" and "small" pertain to the small establishment models.